

Design Portfolio

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Chapter 1 – Introducing Homify

1.1 Introduction

Digital platforms have become the inevitable channel in the current ecommerce sector, with mobile applications playing a major role in how users interact and make purchasing decisions (Jain & Purandare, 2021). In today's world, consumers increasingly prefer online purchasing over visiting physical stores, thus avoiding the inconvenience of product transport and shipping. Mobile applications address this need in the most efficient way by helping users to browse, select, and order products from the comfort of their own home. However, companies that sell home goods still have their own challenges when it comes to online shopping. Customers typically have trouble picturing how a product will look or fit in their environment, and they don't get the same sensory feedback they would get in a real store (Lu & Smith, 2008). This report shows the design portfolio for Homify, a mobile app for e-shopping that focuses on home and furnishing needs. Homify aims to improve the online shopping experience by applying key principles of Human-Computer Interaction (HCI) to enhance user experience (UX) by reducing purchase risk and improving overall efficiency. The visual design of Homify emphasizes clarity and aesthetic simplicity, which follows the principle of Aesthetics and Minimalism (Pereyra, 2023). To create a visual trust and reduce cognitive load, the interface is built in a clean and well-structured way, supported by a warm color palette (Miller, 1956). Also, to meet the needs of users operating under time constraints, Homify integrates advanced features such as Augmented Reality (AR) visualization, biometric/NFC login, collaborative shopping, live delivery, and Google Calendar synchronization. These functionalities provide a smooth flow throughout the user journey right from product discovery to secure payment and product delivery. The design approach adheres to an Iterative Design methodology backed with systematic assessment and improvements in accordance with changing user expectations (Sharp et al., 2007). This document presents the design rationale and validation for the Homify mobile application prototype. It explains how design decisions, functional requirements, and visuals align with established Interaction Design (ID) and UX principles to create an intuitive and engaging system. It also showcases the application of HCI concepts, supported by academic references, to facilitate the design usability, customer satisfaction, and overall efficiency.

Chapter 2 – Persona and Scenario

2.1 Persona 1: Alex

Persona 1 — Alex	
	Name: Alex Grander Type of Persona/User: Senior Manager Age: 30 Occupation: Interior Designer
Other relevant demographics	Recently moved to a new city. Works in a design-heavy role but avoids unnecessary screen time due to migraines. His work schedule is busy, leaving little time for errands or complex digital tasks.
Likes	• Minimalist and well-structured interfaces • Fewer but meaningful features • Quick shortcuts (scan-to-shop, one-click payment) • Aesthetic yet practical product recommendations
Dislikes	• Cluttered and pop-up heavy interfaces • Long login processes and repeated OTPs • Managing different apps for purchase, payment and tracking
Stressors & Motivations	Gets overwhelmed when tasks build up. Sensitive to long screen exposure. Motivated by simplicity, clarity and tools that help him maintain an aesthetically pleasing environment with minimal effort.
Pain Points	• Forgets passwords and dislikes repeated logins • Navigation becomes tiring due to migraines • Delivery updates are missed due to a busy schedule • Dislikes switching between multiple apps to complete one task

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Persona 1 — Alex (continued)	
Needs & Expectations	• Fast login using biometrics or NFC • Clean interface with minimal steps • One platform to browse, purchase, track and pay • Ability to scan home space for design suggestions • Shared cart and split payment for collaborative purchases
Goals & Tasks	• Furnish home quickly, aesthetically and with minimal effort • Reduce screen strain while still completing essential tasks • Coordinate purchases and deliveries with flatmate without confusion

2.2 Persona 2: Priya

Persona 2 — Priya	
	Name: Priya Marino Type of Persona/User: Junior Engineer Age: 24 Occupation: Software Engineer
Other relevant demographics	Recently started a job in a new city. Lives independently, follows hybrid working, and relies heavily on her smartphone for shopping, payments and daily tasks. Values comfort and a personalised living space.
Likes	• Smart digital tools and all-in-one applications • Personalised suggestions and clean UI • Cozy interiors and multi-purpose furniture • Cashback offers and fast delivery
Dislikes	• Physical store visits and long queues • Apps that load slowly or crash • Complicated returns or unclear product information
Stressors & Motivations	Feels unsettled until her surroundings feel familiar and organised. Motivated by visually pleasing spaces that keep her calm and productive while working from home.
Pain Points	Relocation pressure combined with fragmented apps makes her worry whether she can furnish her home the way she wants. She dislikes having to use multiple apps for purchases, tracking and payments.
Needs & Expectations	• Single app to browse, buy, pay and track orders • Ability to preview items in her room • Clear images, details and accurate delivery updates • Fast checkout and smooth digital payments
Goals & Tasks	• Set up her apartment quickly to feel personal and comforting • Compare products easily and stay within budget • Recreate the comforting atmosphere of her previous home city

2.3 Scenario

Alex, an interior designer, and Priya, a software engineer, both move to a new city for work, and they end up sharing a two-bedroom apartment by coincidence. The space has considerable potential. It is bright, airy, and well-structured, but at the moment, it feels empty and disconnected rather than like a real home. Alex immediately envisions how to transform it into a minimal and aesthetically balanced home, but his demanding schedule and recurring migraines make it difficult to spend long hours browsing online stores or visiting furniture shops. The digital platforms he tries are cluttered, require repeated logins, and involve complex navigation.

In contrast, Priya prefers getting everything done through her phone, from groceries to payments, and relies on online furniture apps; however, she quickly realises the process is fragmented where one app provides appealing designs but no AR preview, another offers tracking but has a poor return system, and constantly switching between apps becomes frustrating.

When Alex and Priya try to furnish the apartment together, issues arise as they share screenshots, links and payments across different platforms, which leads to duplicate orders, missed delivery updates and budget confusion. What should have been an exciting decorating process turns into a time-consuming and disorganised experience.

Both of them wish for a single solution where they can browse intelligently recommended products, visualize furniture using AR, share a cart in real-time, split payments, schedule deliveries based on availability and track orders while communicating with delivery partners, all in one place. This shared frustration becomes the foundation for designing a seamless, collaborative and user-centred home furnishing application.

2.4 Rationale for Personas and Scenario using the PACT Model

The personas and scenario were structured using the PACT (People, Activities, Context, Technology) framework (Benyon et al., 2005) to ensure practical relevance and alignment with UX principles.

People: Alex represents users who prioritize design, have limited free time, and prefer minimal interaction with technology due to health or workload. Priya represents users who rely heavily on smartphones and digital tools. She values convenience, personalization and speed when completing tasks.

Activities: Both personas perform similar tasks like browsing, visualizing products, managing payments, coordinating deliveries and reviewing items. Their difficulties include switching between multiple applications, not having the right tools to collaborate, and difficulty in visualizing how the products will look in the real world.

Context: The scenario of two people sharing an apartment creates situations where both individual and shared decisions are needed. This makes it important to support features like shared carts, split payments, delivery planning and avoiding repeated or accidental purchases.

Technology: The scenario justifies integrating augmented reality for product visualization, a shared cart that updates in real time, an option for split payments, calendar-based delivery scheduling, live order tracking with a built-in chat, and biometric/NFC login to support real-time collaboration and reduce cognitive load.

2.5 User Requirements

1. Users should be able to log in using traditional method, biometric authentication, or NFC/membership card tap.
2. The home page must be clean, visually minimal, and emphasize primary features through clear content hierarchy.
3. Users should be able to scan their environment to receive product suggestions that match their interior setting.
4. Users should be able to view how products would appear in their real space using augmented reality.
5. Users must be able to add, remove or modify items in the cart easily. A shared cart feature should allow users to invite others (via email, phone number or link) to view, add or remove products collaboratively in real time.
6. Users must be able to edit delivery details and pick a date based on their google calendar schedule for delivery.
7. Users should have access to multiple payment options including debit/credit cards, Google Pay, PayPal, NFC/tap-to-pay and group wallet.
8. Users must be able to initiate split payments for shared purchases directly within the application.
9. Users should have access to live order tracking and an integrated chat interface to communicate with delivery personnel and other shared recipients.
10. Users must be able to submit, update or modify product reviews at different stages of ownership through a structured review timeline.

Chapter 3 – Features and Design solutions

3.1 Walkthrough of 10 Prototype Screens

The following sections provide a brief walkthrough and rationale for the 10 selected screens.

3.1.1 Screen 1: Welcome Back! (Login)

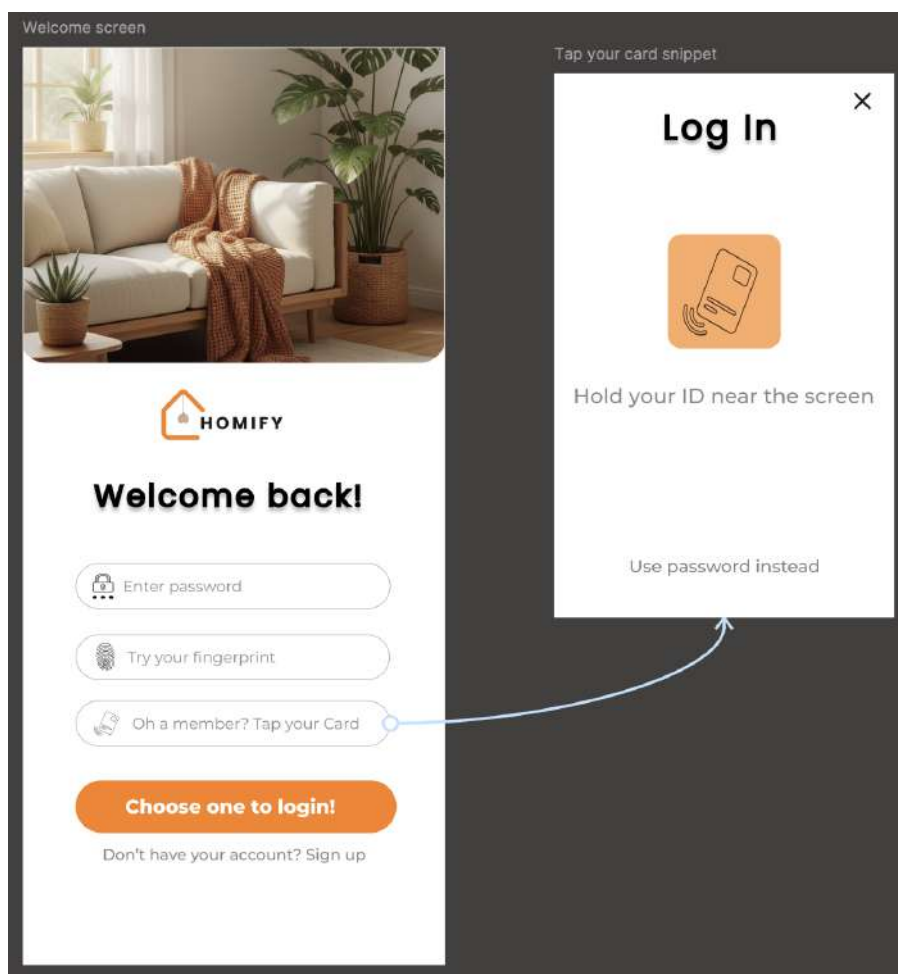


Figure 3.1: Welcome screen with a snippet of tap a card feature

This is the main access point of the app which is designed with a focus on both security and user control. It follows User Control and Freedom (Nielsen, 2005) by offering users multiple ways to login: using a password, fingerprint or through the advanced Tap Membership ID/NFC option. In particular, the NFC method makes the login process quick and effortless (Prasad et al., 2022) using the principle of Efficiency of Use. Moreover,

the prompt 'Don't have an account? Sign up' uses clear, straightforward language to help users register if they are new. As per the snippet in the Figure 3.1, if the NFC card tap method isn't successful due to some issue, the users can log in by entering their password.

3.1.2 Screen 2: Home landing page

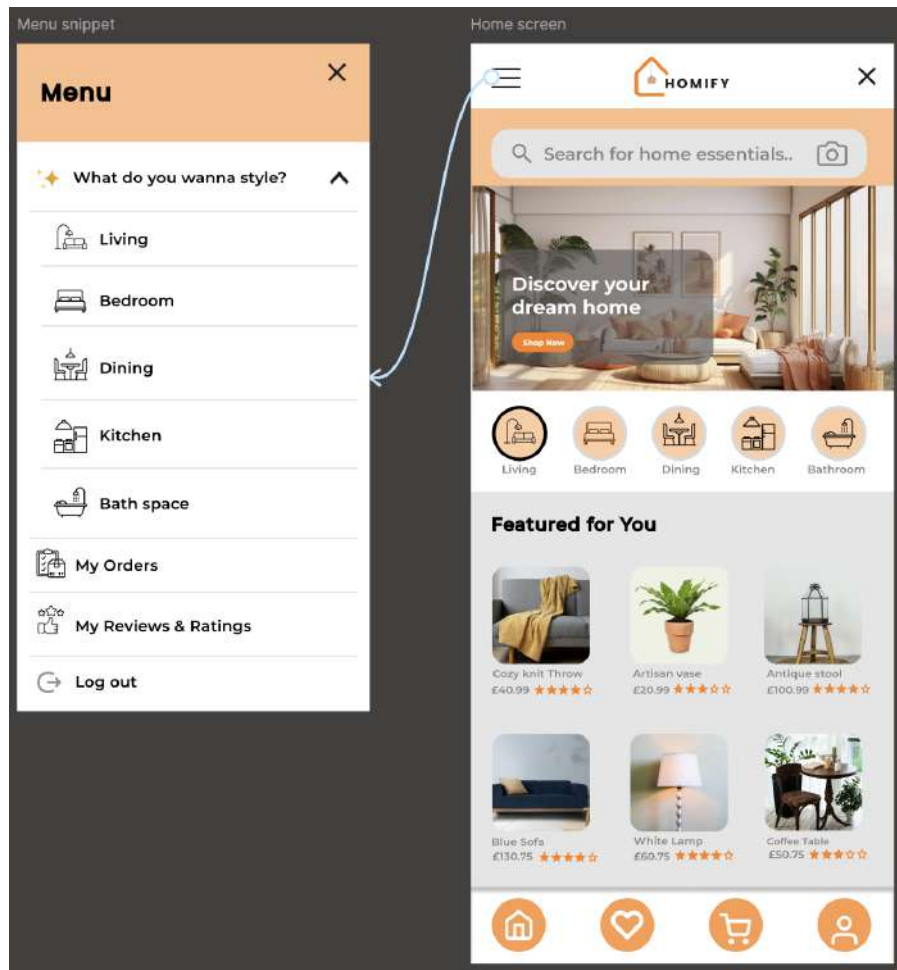


Figure 3.2: Home landing page screen with a snippet of main menu

The Home page is designed to help users find their way around easily and quickly by adhering to the principle of Visibility of Status (Norman, 2013). The interface balances aesthetics and usability to prevent Information Overload (Miller, 1956). The design uses a Clear Visual Hierarchy via a hero banner and chunked categories (e.g., Living, Dining etc.). Hence, core functions can be found without cognitive friction. Users can instantly search for products by typing keywords into the search field or by clicking on the adjacent 'Scan to Style' camera icon to find products based on their surroundings. The design has high-contrast text and a persistent bottom navigation bar for overall accessibility and continuous user control (Jain & Purandare, 2021). This bar maintains constant access to primary actions such as the user profile, wishlist, and cart which helps user efficiency (Gatsou et al., 2012). In addition, a hamburger menu is located on the top left corner, which expands to display the whole navigation panel and provides shortcuts links. Users

can dismiss this overlay easily by clicking the close button, providing complete control and freedom during their browsing session (Nielsen, 2005).

3.1.3 Screen 3: Scan to Style

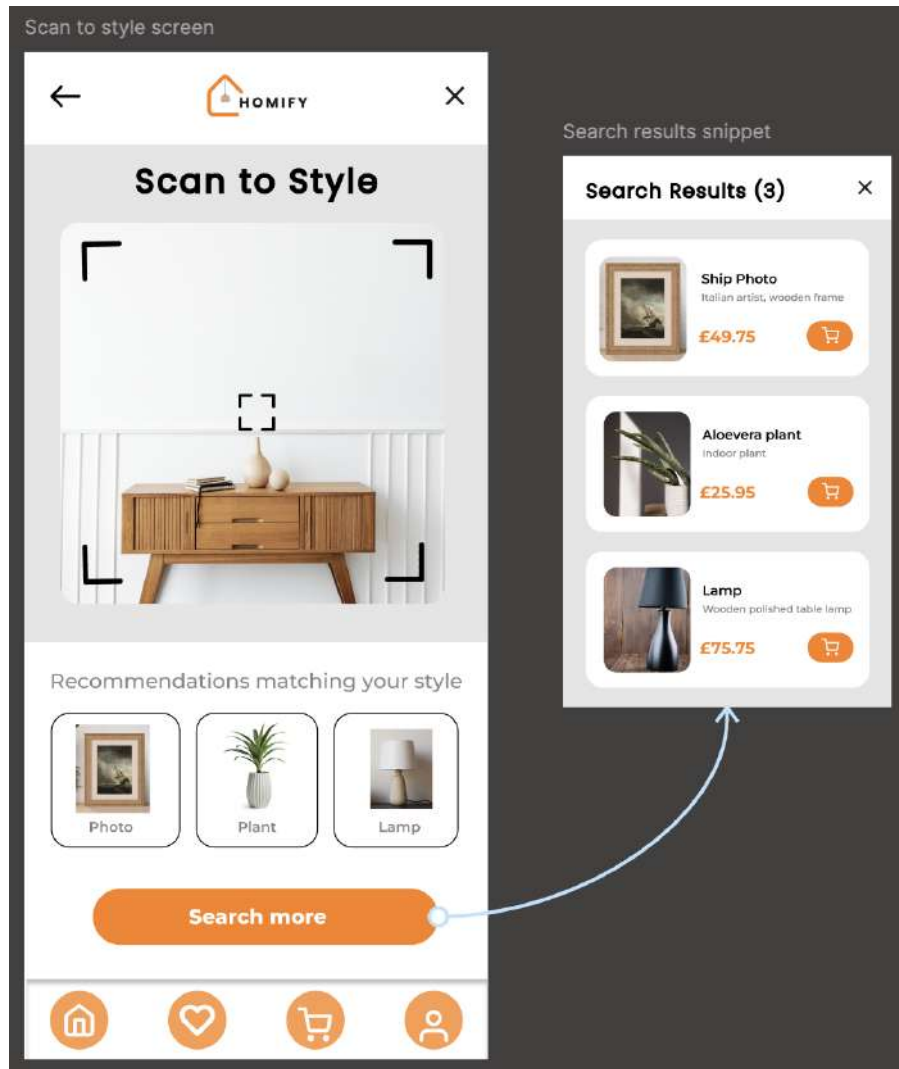


Figure 3.3: Scan to style screen with a snippet of search results

This screen initiates the advanced environment scan flow using AR technology. It follows the user-centred design principle by allowing the user to take a picture of an existing space (e.g., dining area). The system then provides context-aware product suggestions (e.g., accent elements like a lamp) to enhance the user's existing style. It gives feedback (Norman, 2013) by showing product thumbnails results before the user initiates the final search, which takes him/her to the appropriate search results. Here, the use of icons supports efficient visual communication (Gatsou et al., 2012). Users can add products directly from the search results to their cart.

3.1.4 Screen 4: Product Details

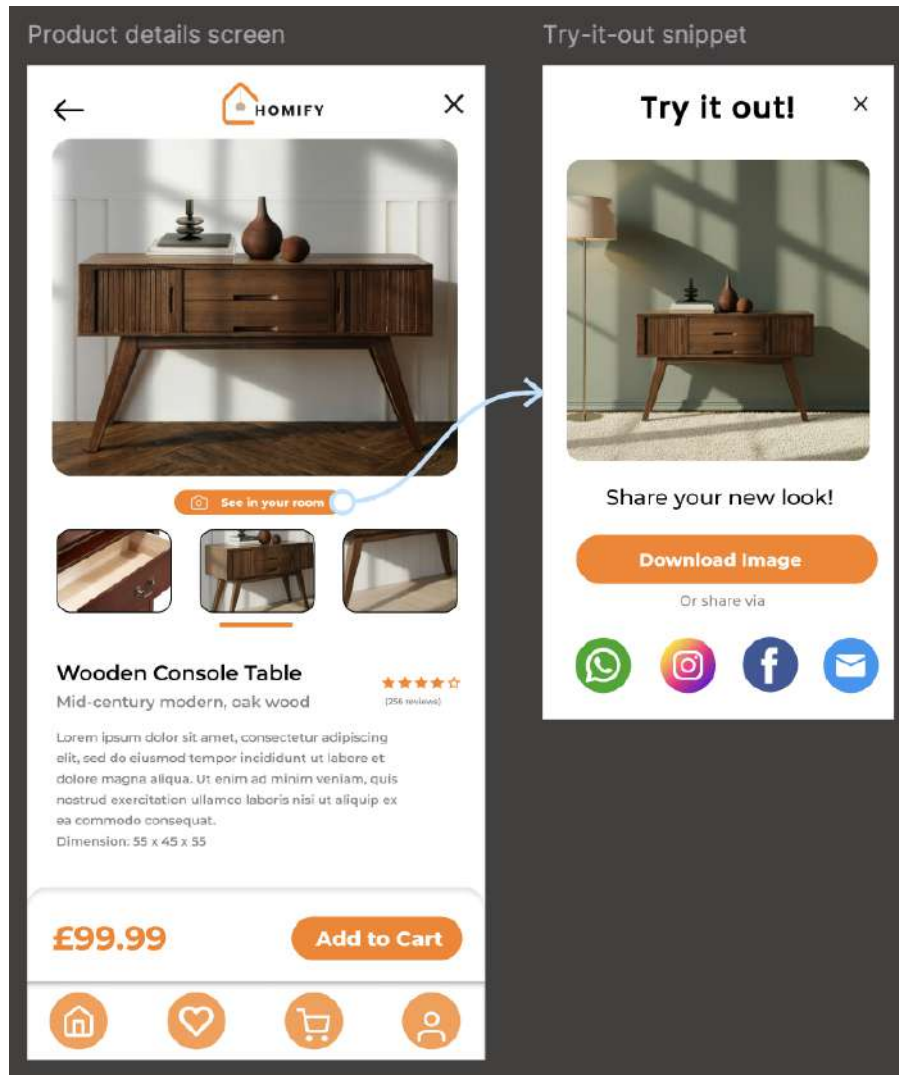


Figure 3.4: Product details screen with a snippet of Try it out feature

This screen displays the product image with its details. The overall structure uses chunking and clear typography to present price, rating stars for Recognition rather than Recall (Nielsen, 2005), and product information. This directly supports the user's need for detailed visual information before a purchase (Lu & Smith, 2008). The advanced call-to-action (CTA), 'See in your room' is placed directly under the image, adhering to the principle of Affordance (Norman, 2013). Upon clicking the 'See in Your Room' CTA, the user is redirected to the 'Try It Out!' screen (snippet of which is presented in Figure 3.5), where they can visualize how the product would appear in their own space. Later, the user can download the generated image to their device or share the new look with friends/peers via WhatsApp, Instagram, Facebook, or email.

3.1.5 Screen 5: Cart

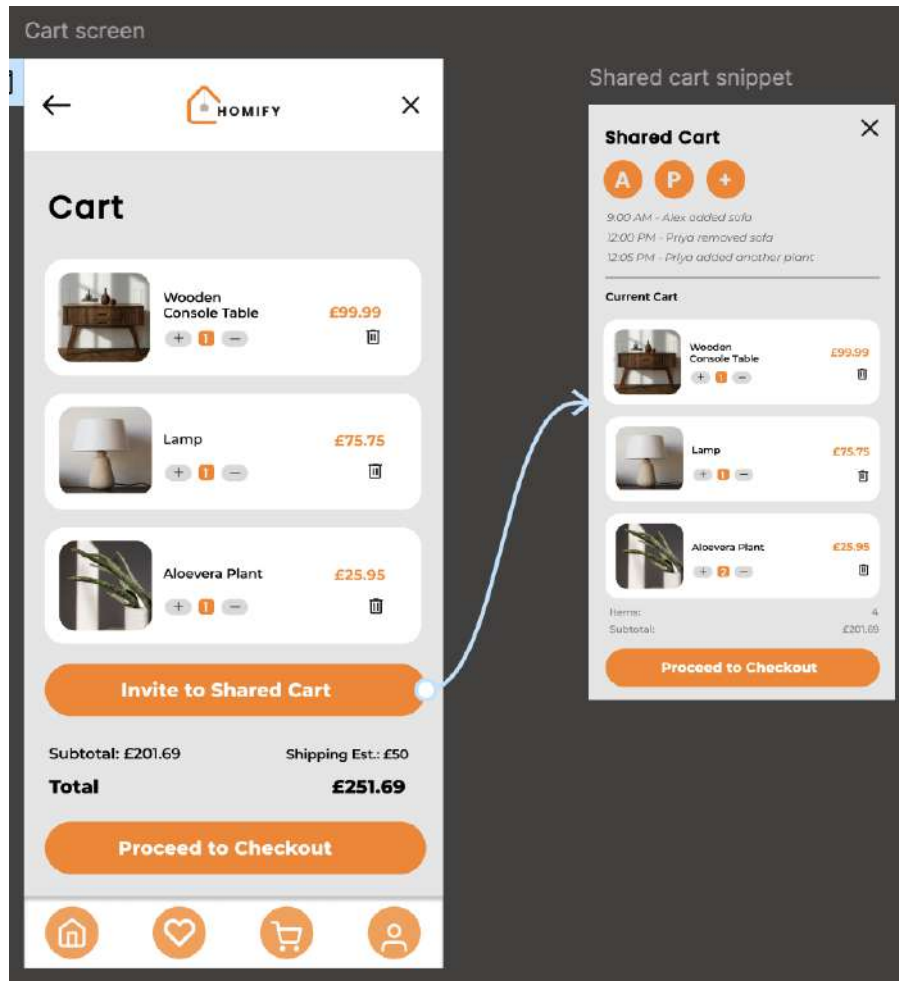


Figure 3.5: Cart screen with a snippet of shared cart feature

This screen displays a basic ecommerce function with an advanced feature prompt. Product details are clearly listed with large thumbnails and quantity control buttons, which adhere to Touch Target Size principle as stated in Fitts' Law (Fitts, 1954). The advanced CTA 'Invite to Shared Cart' is positioned prominently above the subtotal, which helps to provide the collaborative option visible without forcing the user to use it, and this positioning is as per the Progressive Disclosure principle of Interaction Design (ID) and usability engineering. After reviewing the items in the cart, the user can proceed directly to checkout or choose the shared cart feature. As shown in Figure 3.5, the shared cart feature allows users to add multiple members to collaborate on a purchase. It displays member avatars and a Visual Narrative of recent actions (e.g., 'Alex added sofa', 'Priya removed plant'), providing feedback (Norman, 2013) and visibility of status (Nielsen, 2005) on the group's activity, along with timings. This supports a user-centred activity of co-shopping. Once the group reaches a consensus, the selected items in the cart can be pushed to the checkout stage.

3.1.6 Screen 6: Checkout

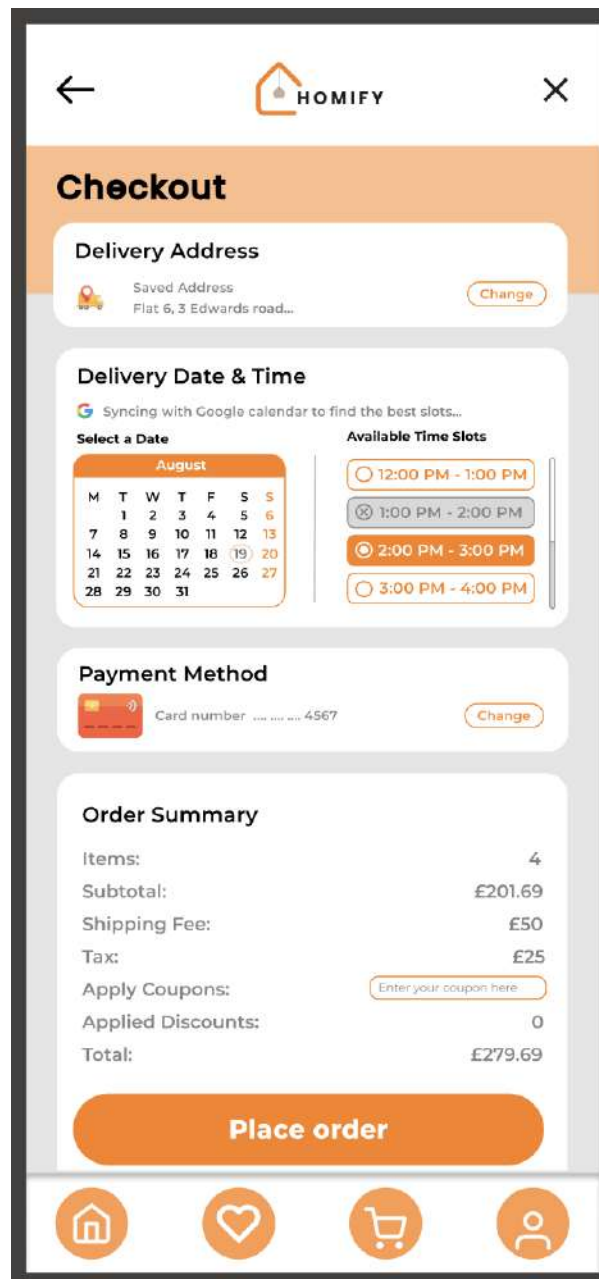


Figure 3.6: Checkout screen

On this screen, users can verify their delivery address and edit it if necessary by selecting the 'Change' CTA. The delivery date & time picker is in sync with the user's Google Calendar. Clear visual cues are used to avoid errors (Nielsen, 2005). Unavailable slots are shown in grey, and available slots are shown in orange, maintaining consistent color coding to convey system status. The selected time slot will be painted in orange. Then, users can select their preferred payment method. The 'Change' CTA, located next to the user's default payment allows to select their desired mode of payment. Finally, users can check their order details, use any coupons, and hit 'Place Order' to complete their purchase.

3.1.7 Screen 7: Payment

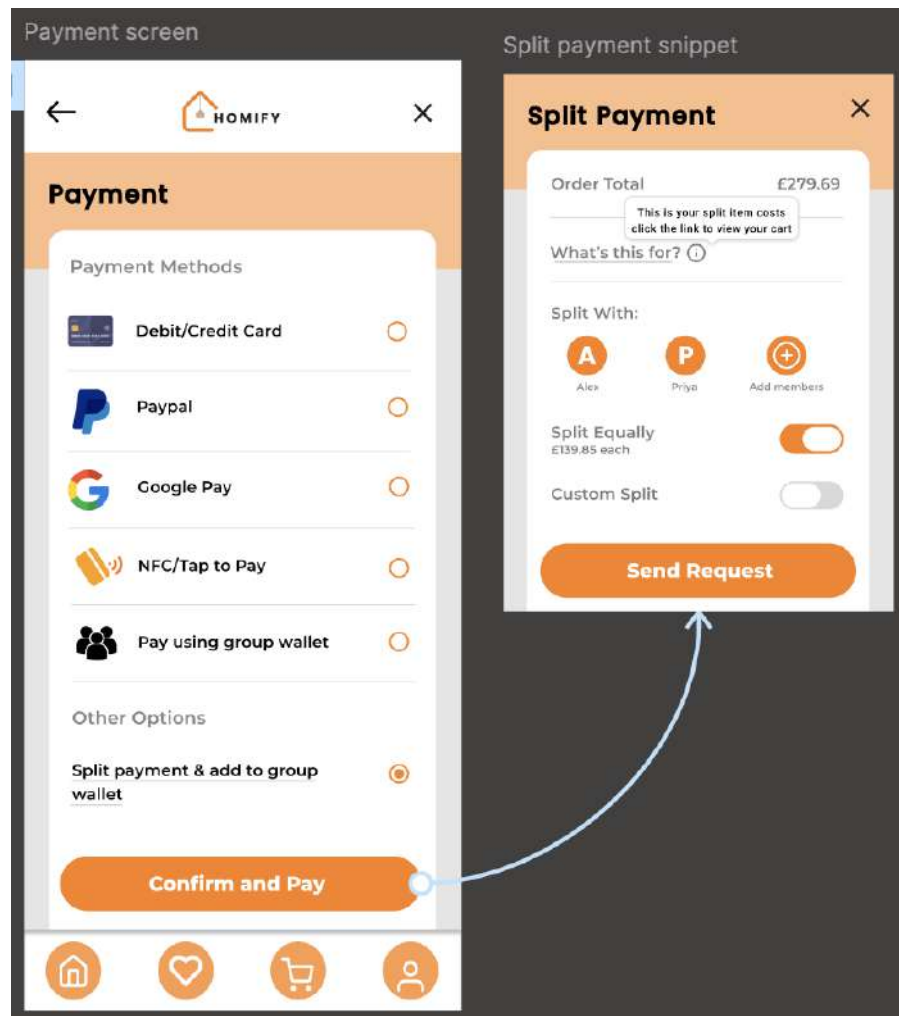


Figure 3.7: Payment screen with a snippet of shared cart feature

When the user wants to change the payment mode, the payment screen shows up. It supports payments using a debit/credit card, PayPal, Google Pay, Card Tap/NFC or Group Wallet. Users can also opt to split payments by adding funds to the group wallet and paying together. Users can proceed further by clicking the 'Confirm and Pay' CTA once they have chosen how they want to pay. The Split Payment screen as per the snippet in Figure 3.7 manages the complexity of shared financial transactions. It provides constant visibility of status (Nielsen, 2005) via the order total and the amount split. The design is user-centred as it allows the user to select members and manually adjust amounts either by splitting equally or by custom split, which relates to the principle of flexibility & efficiency (Nielsen, 2005). The 'Send Request' CTA is a clear prompt for the final action.

3.1.8 Screen 8: Order Confirmation

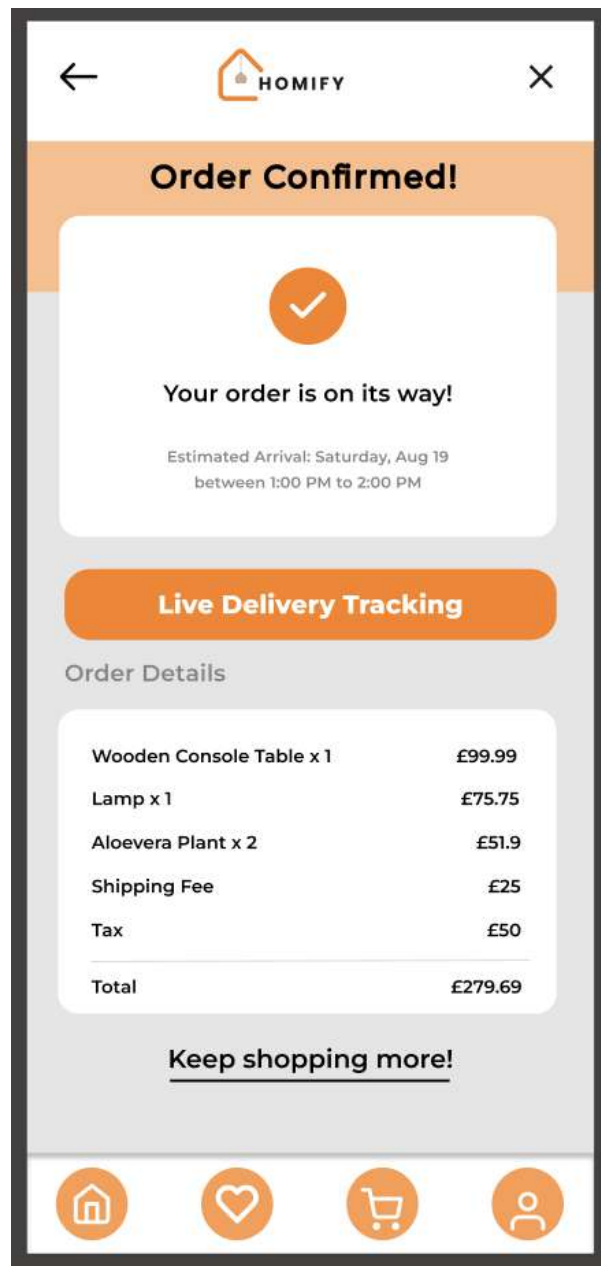


Figure 3.8: Order confirmation screen

This screen provides closure and immediate confidence. The checkmark and 'Your order is on its way!' provide positive feedback (Norman, 2013), aesthetics & minimalism (Pereyra, 2023). The primary CTA, 'Live Delivery Tracking' offers clear affordance (Norman, 2013) and guides the user to the next stage of the post-purchase process. The interface includes a 'Keep Shopping More' link designed to encourage further product exploration and increase user engagement.

3.1.9 Screen 9: Live Delivery

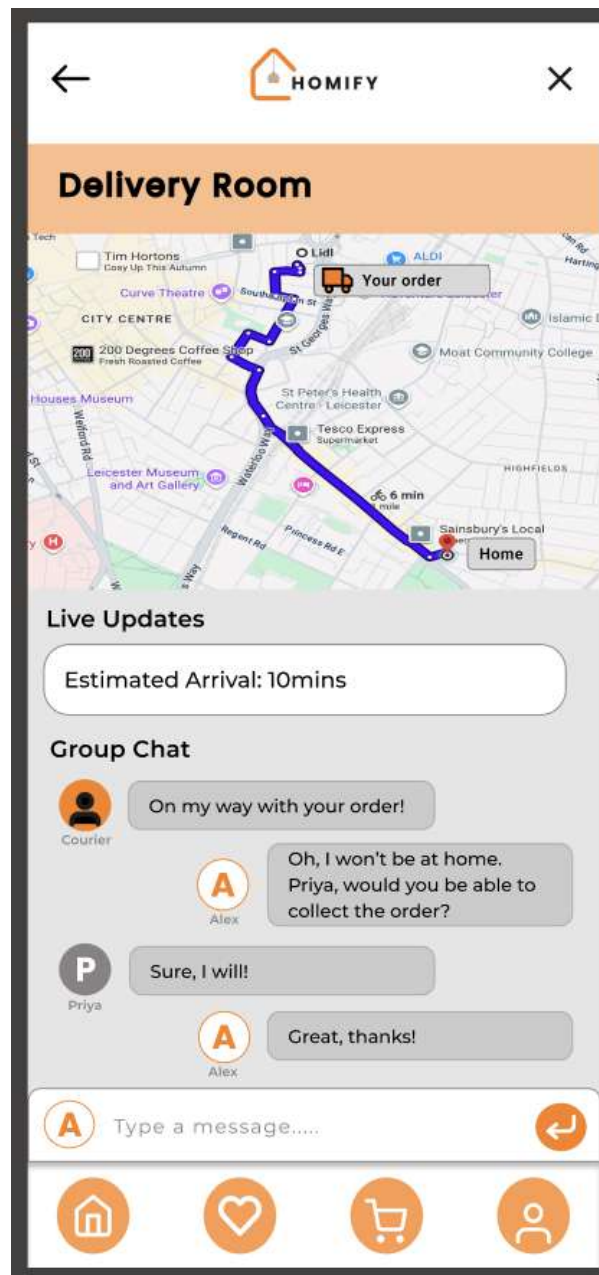


Figure 3.9: Live delivery screen

This screen emphasizes trust and transparency to users. It features Google Maps and real-time ETA (Estimated Time of Arrival) for full visibility(Norman, 2013). The integrated group chat between the courier and recipients (Alex and Priya) is an advanced feature that reinforces error recognition and recovery (Nielsen, 2005). It also allows for direct and timely communication to resolve delivery issues and ensure the handover is smooth even if the primary user isn't available(Vargo & Lusch, 2004).

3.1.10 Screen 10: Product reviews

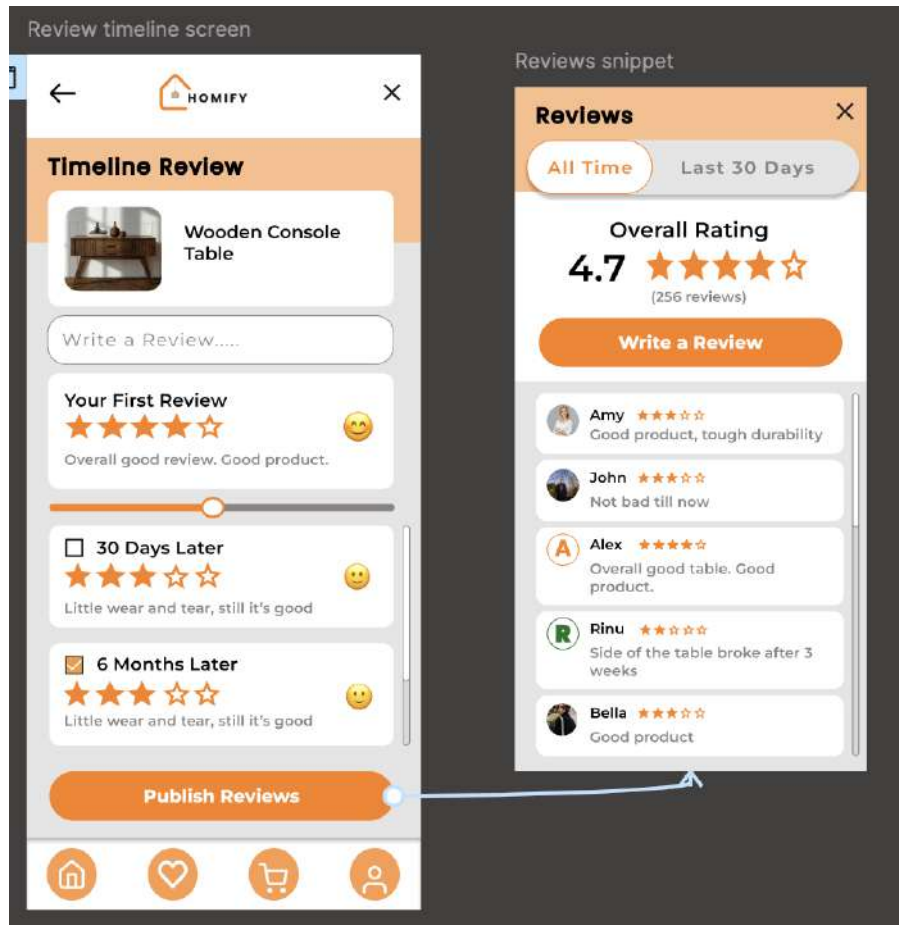


Figure 3.10: Product reviews snippet with timeline review screen

This screen lets users create and manage product reviews anytime, it's accessible via the 'My Reviews and Ratings' menu on the homepage. The design uses a clear visual narrative that makes the review process more like a game and lets users to give feedback at important points in their ownership, such as 'Your First Review', '30 Days Later', etc. Every milestone includes a selectable checkbox. Clicking the 'Publish Reviews' CTA displays only the selected reviews to other users. This staggered approach uses iterative design concepts to collect user feedback regarding product performance and durability, which is crucial for expensive items like furniture. The timeline slider promotes visibility of system status (Nielsen, 2005) by indicating users of the time period their reviews correspond to. The snippet from Figure 3.10 is the reviews display interface, showing what other customers see. It adheres to Aesthetics & Minimalism (Pereyra, 2023) and Clear Hierarchy (Nielsen, 2005) by presenting the most critical information first: the Overall Rating, the star average, and the total review count. The tabbed structure ('All Time', 'Last 30 Days') adopts Progressive Disclosure to mitigate Information Overload (Miller, 1956) which allows users to filter content and experience a clean view.

Chapter 4 – Conclusion

The prototype of Homify mobile application successfully converts strategic ID and UX principles into a feature-rich, low-friction ecommerce experience. This design portfolio satisfies all essential user needs and functional requirements, including secure login and post-purchase delivery coordination. The aesthetic appeal and commitment to Aesthetics and Minimalism (Pereyra, 2023) establish immediate visual trust, which is crucial for premium purchases. Homify is designed to mitigate the two primary challenges of mobile ecommerce: Complexity and Purchase Risk. Design decisions completely adhere to principles of Clear Hierarchy and Progressive Disclosure (Miller, 1956), thus preventing the proliferation of advanced features (AR, Split Payment) from leading to Information Overload. Notably, this design contributes to current literature by offering practical answers to well-documented problems. By implementing AR visualization (Figure 3.4), Homify directly tackles the sensory and dimensional uncertainty present in online furniture shopping, one of the limitations identified in early ecommerce studies (Lu & Smith, 2008). Furthermore, the inclusion of collaborative technologies like Split Payment (Snippet from Figure 3.7) and Shared Cart (Figure 3.5) match with advanced theories on co-creation of value (Vargo & Lusch, 2004). All these are transforming the shopping activity from a singular, isolated task into an efficient social interaction. The strategic use of Platform Integration, such as the Google Calendar integration for delivery scheduling (Figure 3.6) and the extensive live delivery room (Figure 3.9), supports the theory that leveraging platform resources is key to achieving superior efficiency and competitive advantage (Gallino & Moreno, 2013). To summarize, Homify is more than just an ecommerce app; it is a solution that re-imagines the relationship between the user, product, and platform. It is the best solution that sets a new benchmark for ID and UX in the home furnishings sector.

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